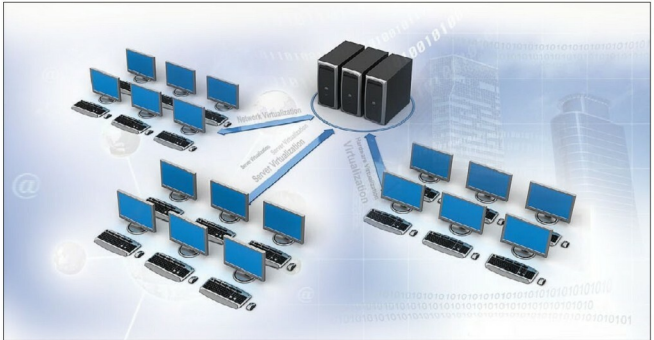


Mininet – An Emulator for Prototyping Large Network Topologies on a Single Machine

Mininet is a useful tool for teaching, development and research. With it, a realistic virtual network, running a real kernel switch and application code, can be set up in a few seconds on a single machine, either virtual or native. It is actively developed and supported.



Emulation refers to the running of unchanged code on virtual hardware on the top of the physical host, interactively. It is handy, practical and low cost. It comes with certain restrictions, though, like slower speeds compared to running the same code on a hardware test-bed which is fast and accurate, but expensive. While a simulator requires code modifications and is slow as well.

Network virtualisation

Computer networks are constantly evolving. The next generation networks have to be scalable, programmable and flexible enough to adapt to innovative ideas. Researchers continue to work on providing innovative solutions or modifications to the current physical network infrastructure. The existing networks are not capable of performing as experimental test beds. So, there is a need for new customised test-beds for experimentation. What is required are new protocols which do not affect existing network infrastructure and operations and this need is addressed by the virtual network infrastructure.

Virtualisation refers to the process of framing a virtual environment to execute experimental work. It could be

virtualisation of hardware, platforms, servers or a network. We can create such an experimental test-bed using the Mininet emulator.

Introducing Mininet

Mininet is a network emulator that enables the creation of a network of virtual hosts, switches, controllers, and links. Mininet hosts standard Linux network software, and its switches support OpenFlow, a software defined network (SDN) for highly flexible custom routing.

It constructs a virtual network that appears to be a real physical network. You can create a network topology, simulate it and implement the various network performance parameters such as bandwidth, latency, packet loss, etc, with Mininet, using simple code. You can create the virtual network on a single machine (a VM, the cloud or a native machine).

Mininet permits the creation of multiple nodes (hosts, switches or controllers), enabling a big network to be simulated on a single PC. This is very useful in experimenting with various topologies and different controllers, for different network scenarios.